Roy (Yuanxi) Cheng

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Education

University of California, Los Angeles (UCLA) Sep 2023 – Mar 2027

Bachelor of Science in Statistics and Data Science, Minor in Data Science Engineering *GPA: 3.75*

- **Awards & Programs:** Dean's Honor's List, Persolv AI Intensive, BCG Case Comp Semi-finalist
- **Certificates:** Data Science & SQL in Python, EDA in R, Data Manipulation with dplyr, Data Visualization with ggplot2
- **Relevant Coursework:** Machine Learning, Neural Networks & Deep Learning, Probability & Statistical Inference, Data Mining, Data Analysis & Regression, NLP, Optimization, Monte Carlo, Databases, Data Structures & Algorithms

Projects and Research

Biomedical Tissue Imaging | *Python, PyTorch, YOLOv8, Pandas, NumPy, Matplotlib, OpenCV*

- Published article analyzing CNN-based segmentation models with AP metrics to evaluate cytoplasm boundary precision
- Engineered data pipeline to convert 4k+ CytoNuke annotations into COCO to allow training across 3 model architectures
- Trained YOLOv8 instance segmentation in PyTorch, improving cytoplasm AP50 by 5% over the Cyto-RCNN baseline

COVID Pandemic Effects on Secondary Education | *Python, Pandas, Tableau, Excel, Matplotlib*

- Cleaned 100k+ student records by merging multi-level datasets, enabling accurate longitudinal analysis in Tableau
- Aggregated 4 years of test scores into time series visualizations, revealing a 15% average decline in student performance
- Created interactive Tableau map for research website, linking geographic data to score changes across 50 U.S. states

Adrenaline and Cognitive Fatigue Effects on Athletic Performance | *R, ggplot2, tidyverse, dplyr*

- Designed 2^3 factorial experiment with 96 participants across 8 conditions, blocking by age and gender to reduce variance
- Conducted three-way ANOVA and Tukey post hoc tests in R, revealing no significant interactions at $\alpha = 0.05$
- Verified model assumptions with log transformation and ggplot2 diagnostics, confirming normality and homoscedasticity

Work and Leadership Experience

Capital One McLean, Virginia

Product Management and Analysis Intern *Jun 2026 – Present*

- Analyze corporate workforce data with SQL and Python, identifying effects and factors tied to successful org transitions
- Architect Python pipeline through PII anonymization, BGE filtering, and FAISS retrieval to an LLM DWA scoring agent
- Build custom LLM client with sliding-window rate limiting and retry-aware backoff for multi-hour pipeline throughput

Data Science Union Los Angeles, California

External Vice President *Dec 2025 – Present*

- Directed client partnerships, marketing, and professional development initiatives to expand opportunities across members
- Managed 3 directors and coordinated executive priorities, ensuring seamless execution of quarterly and annual objectives
- Cultivated industry and alumni partnerships to secure client data science projects and expand career programming

Kohl's Los Angeles, California

Project Manager and Data Analyst *Jun 2025 – Mar 2026*

- Processed 8.5M+ logistics records in BigQuery, building SQL validation to remove duplicates and leakage-prone fields
- Developed two-stage XGBoost forecasting system simulating live updates, cutting shipment arrival MAE by 83%
- Engineered feature pipelines for route, carrier, and seasonality in Vertex AI Workbench for transit time forecasting

ACTUM Digital Prague, Czech Republic

Data Analyst and AI Intern *Jun 2025 – Aug 2025*

- Built agentic Python pipelines to filter documents, generate metadata, and embed content into Milvus for AI retrieval
- Authored prompting and integrated MCP server to create AI assistant, co-presenting live demo to ACTUM executives
- Automated extraction from 600+ unstructured PDFs with Python, consolidating results into a centralized dataset

Million Dollar Baby Los Angeles, California

Data Analysis Consultant *Apr 2025 – Jun 2025*

- Analyzed $10M+ ad spend data with SQL to extract insights that informed growth strategy and investment decisions
- Presented CAC and RoAS assessments across attribution windows to guide stakeholders on marketing spend allocation
- Built Excel dashboard of 20+ marketing KPIs, giving brand managers a recurring view of acquisition and return trends

Technical Skills

Languages: Java, Python, C++, SQL, R, JavaScript
Libraries/Frameworks: PyTorch, pandas, NumPy, scikit-learn, Matplotlib, TensorFlow, XGBoost, LangChain, MLflow
Tools & Platforms: AWS, GCP, BigQuery, Snowflake, Git, GitHub, Vertex AI, Jupyter Notebooks